| Problem Chosen | 2020 | Team Control Number |
|---|---|---|
| C | MCM/ICM | 7391856 |
| | Summary Sheet | |

# Product Investigation and Reputation Monitoring

## Summary

Sunshine needs to decide which product problems to investigate and when worsening reviews warrant a response. We analyze the 32,024 supplied reviews using rating distributions, keyword-based text measures, helpfulness interpreted alongside voting exposure, and a calendar-based reputation model. These measures describe review behavior; the files do not measure sales.

One–two-star reviews account for 31.8% of the microwave file, compared with 14.6% of the dryer file and 11.3% of the pacifier file. Product scope also matters: a conservative title-defined pacifier cohort retains 10,475 of 18,939 rows and has a low-star share of 9.1%. In the electrical categories, reliability mentions are associated with within-family low-star differences of 38.07 and 41.10 percentage points between mentioning and non-mentioning reviews; both associations remain positive after every one-family deletion. This evidence makes durability investigation our strongest recommendation. Acoustic expressions require a different interpretation: separating noise from quiet reveals a dryer contrast obscured by their merged topic, whereas the microwave comparison has only one eligible family. The pooled pacifier material association largely disappears within families and does not support a universal material preference.

Our dashboard combines a rolling ninety-day low-star fraction, a Beta posterior interval, issue counts, verified-review strata and vote coverage. Calendar comparisons reveal composition effects: pooled reputation trends can hide changes within existing families. Cumulative helpful votes are useful for retrospective review reception but excluded from forecasting because their historical availability is unknown.

To compare rating-only, text-only and combined warning measures, we train through 2013, select on 2014, and test complete ninety-day windows in 2015. In the primary reviewer-rich test of 3901 future reviews, the combined model has log loss 0.32368 versus 0.32402 for ratings alone. Its tiny incremental gain has a product-cluster interval spanning zero. Scoring all positive future windows eligible from past information raises its loss to 0.33104, exposing harder sparse-review outcomes. We recommend a simple rating warning baseline with text for interpretable triage, pending prospective calibration.

Product/time-adjusted review-flow associations do not clearly differ from zero. More fundamentally, missing purchases, impressions and observed rating exposure prevent identification of whether seeing low stars *causes* reviewing. We give a constructive counterexample and a practical data requirement rather than infer causality from sequence. The resulting strategy gives product-investigation priorities and a worked risk-policy decision, with explicit sample, time and measurement limits.

**Keywords:** review reputation; product reliability; temporal validation; text measurement; causal identification

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# Letter to the Marketing Director

Dear Marketing Director,

The most defensible first step is to prioritize reliability investigation for the new hair dryer and microwave. In the supplied reviews, 51.3% of target-title dryer reviews mentioning breakage or stoppage and 80.0% of corresponding microwave reviews carry one or two stars. Within-family comparisons still show roughly 38 and 41 percentage points of excess dissatisfied reviewing. These signals survive exclusion of explicit accessory titles and every one-family deletion in the eligible comparison. They justify examining durability and returned-unit failure modes; they do not yet quantify the sales benefit of redesign.

Keep the dryer easy to handle and quick to dry, and tell microwave buyers the usable capacity and plate fit. Read dryer noise and quiet expressions separately: their merged topic hides different rating relationships. The microwave acoustic comparison is too concentrated in one family for a general design recommendation; favorable capacity discussion does not imply every buyer wants a larger oven. For the pacifier, separate genuine pacifiers from clips, cases and novelty products, publish clear age/size and cleaning information, and test acceptance in the intended use group. A pooled material-word association vanishes within product families, so it should not dictate a universal material choice.

Once products launch, track the last ninety days of ratings with the number of reviews, low-star share, full star distribution and uncertainty interval. Add counts of specific reliability, fit and cleaning complaints, verified-review strata, and vote coverage. Treat a high-probability rise of ten percentage points above a disclosed category reference as an investigation trigger, subject to prior sensitivity and actual text review. Sparse launches require more evidence before a reputation label. Set an acceptable own-product risk tolerance separately: absence of an alert above a poor historical reference does not certify acceptable quality.

We tested combinations of text and stars against future ninety-day reviews. The combined model was nominally best, but its gain over rating-only prediction was small and statistically uncertain; use text to explain and route complaints rather than claim a major predictive improvement. Cumulative helpful votes show how existing reviews are received and must be kept separate from a contemporaneous quality score.

Finally, these records do not show whether a sequence of low ratings incites new reviews. Our adjusted associations are inconclusive, and purchases, impressions and each customer's displayed review history are absent. Review counts can change because sales change or because willingness to review changes. Avoid designing rating campaigns around an unsupported causal story.

We recommend a reliability-first engineering investigation and a transparent, uncertainty-aware review dashboard. This recommendation rests on observed associations and explicit checks, while sales effects, defect incidence and performance on your brand-new listings require additional evidence.

Sincerely,
Team 7391856

# 1 Introduction and study scope

Sunshine plans to introduce a hair dryer, a microwave and a pacifier, and seeks evidence for choosing design priorities and monitoring their reception after launch. The supplied reviews describe ratings, product-feature mentions and other readers' responses to reviews. They do not record sales, profit, page impressions, non-reviewers or the reviews displayed to individual buyers. We therefore distinguish three questions: which features accompany dissatisfied reviewing, whether recent ratings and text predict future low-star reviews, and what the observed sequence can reveal about subsequent review volume. Each question requires a different comparison; evidence for an association does not by itself establish predictive value or a causal response.

Requirement 1 is addressed by Sections 3–4; 2(a) and 2(b) by Section 5; 2(c) by Section 6; 2(d) by Section 7; and 2(e) by Section 4. The preceding letter addresses requirement 3. We use all three supplied files and their glossary [1]. Mathematical formulas below specify our own estimation and comparison choices; all numerical product evidence comes from the supplied records. Study provenance is recorded in the reproducibility appendix and the AI-use report.

Let a review have star rating $r \in \{1, 2, 3, 4, 5\}$ and define the low-star indicator $b = \mathbf{1}\{r \leq 2\}$. Its product family is the supplied `product_parent`; the glossary explicitly permits aggregation by that field. For a set of $n$ reviews, $k = \sum b$, $q = k/n$, and $\bar{r} = \sum r/n$. The denominator for $q$ is *reviews*, never purchases. A higher $q$ indicates a worse observed reviewer reputation under this definition. A commercially failing product could have favorable reviews but insufficient margins or sales; such failure is not measurable here.

# 2 Assumptions and measurement boundaries

1. **Observed reviewers define the estimation population.** The extraction and willingness to review are unknown. We describe these records without assuming random purchasers. This restricts all prevalence statements and prevents conversion of review volume into sales.

2. **Family aggregation follows the supplied identity.** Variants can share a parent while differing in material or design. We therefore compare both pooled and within-family issue associations; family-level alarms require inspecting the actual variant before action.

3. **One–two stars operationalize dissatisfied reviewing.** This is an explicit measurement choice, supported by the supplied scale, rather than a learned definition of market success. A category-level one–three-star summary records descriptive label sensitivity, and verified-only summaries describe reviewer mix. Robustness of downstream forecasts and family issue effects to the one–three-star label is not established.

4. **A category prior stabilizes sparse windows.** Its strength of ten pseudo-reviews is a modeling choice, not an observed fact. We also examine strengths two and thirty. Probabilities from this model are conditional calculations; they are not calibrated guarantees for new Sunshine products.

5. **Text describes mentions.** A transparent dictionary measures words about enthusiasm,

disappointment, and design topics. Local negation reduces some errors but does not resolve sarcasm, quotations or multiple product experiences. A mention is not a certified defect or a safety event.

6. **Historical information must exist at its cutoff.** Stars, text and review dates enter forecasting. Helpful votes are cumulative at extraction and lack a vote date, so they do not enter historical models. Validation covers the same 90-day horizon as the proposed warning model.

7. **Panel adjustment is associative.** Family and calendar effects remove stable listing differences and common time changes only. Exposure, purchases, selective reviewing and unobserved quality remain. We do not assume away these mechanisms to claim that ratings incite reviews.

# 3 Data coverage and reviewer composition

**Table 1.** Complete supplied samples. Low-star means one or two stars.

| Category | Reviews | Families | Mean stars | Low-star | Voted |
|---|---|---|---|---|---|
| Hair dryer | 11,470 | 473 | 4.116 | 14.6% | 37.7% |
| Microwave | 1,615 | 55 | 3.445 | 31.8% | 66.9% |
| Pacifier | 18,939 | 5,432 | 4.305 | 11.3% | 28.0% |

All 32,024 supplied rows have usable dates and integer stars in the stated range; no review ID repeats within its category. Helpful votes are nonnegative and never exceed total votes. We preserve the input files and all valid observations, including extreme votes and unpopular ratings. The earliest dates differ by category, and all files end on August 31, 2015. Inter-category comparisons consequently refer to these unequal extraction samples, not an equal market experiment.

The microwave sample has the lowest mean and the largest low-star fraction (31.8%), compared with 14.6% for hair dryers and 11.3% for pacifiers. Five-star reviews dominate each file. Thus a mean alone hides a substantial dissatisfied tail, particularly for microwave listings; a five-star share alone obscures its size. Retaining the full histogram alongside low-star risk gives an actionable picture without compressing all information into an arbitrary composite score.

## 3.1 Reviewer and listing composition

Purchase and Vine indicators have uppercase and lowercase forms in the supplied files. We retain their original counts and normalize case before grouping. This matters especially in the pacifier file: treating lowercase “y” as unverified would misclassify thousands of records.

Verified microwave reviewers have a much lower low-star fraction than unverified reviewers. This is an association, not evidence that verification improves products or that unverified complaints are false. Different products, review timing, return experiences and reviewer selection could create it. We retain both strata and advise Sunshine to show them separately. Vine

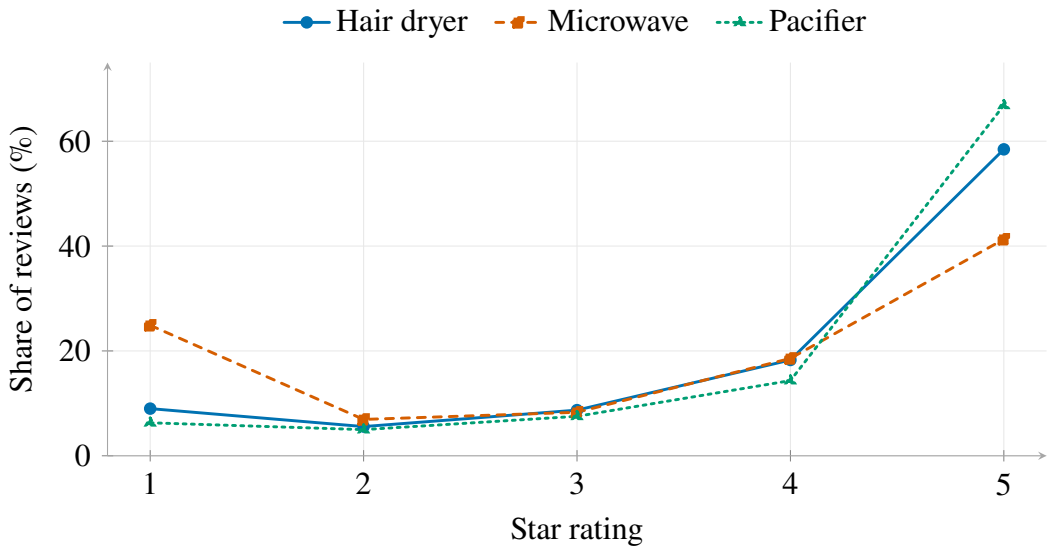



**Figure 1.** Star-rating profiles. Each denominator is its complete supplied category; connecting lines guide comparison of five discrete levels.

**Table 2.** Purchase-verification sensitivity after case normalization.

| Category | Verified *n* | Verified low | Other *n* | Other low |
|---|---|---|---|---|
| Hair dryer | 9811 | 12.7% | 1659 | 25.5% |
| Microwave | 1095 | 15.7% | 520 | 65.8% |
| Pacifier | 16261 | 10.3% | 2678 | 17.2% |

samples are small and more favorable in several categories; their favorable ratings cannot be treated as unbiased launch predictions.

The files also contain surrounding and unrelated products. For example, microwave cavity paint and a magnetron/diode kit appear in the microwave file; pacifier storage boxes, a baby-shower game and a thermometer appear in the pacifier file. Concordances also expose baby cups and video monitors; the hair-dryer file contains some unrelated beauty products. Deleting them silently would change the question's provided samples. Instead, our main results retain every row and a separate sensitivity excludes titles matching declared accessory words.

**Table 3.** Title-exclusion sensitivity; excluded records remain in every main result.

| Category | Excluded $n$ | Retained $n$ | Retained low |
|---|---|---|---|
| Hair dryer | 69 | 11401 | 14.5% |
| Microwave | 136 | 1479 | 34.3% |
| Pacifier | 842 | 18097 | 11.1% |

This transparent exclusion is imperfect: a combined pacifier-and-clip title may be removed, while an unusual accessory without those words may remain. It is a sensitivity, not a manually verified ontology. After exclusion, the low-star fractions for reliability-mention reviews are Hair dryer: 50.9%; Microwave: 79.7%. The reliability signal survives this direct challenge; its exact prevalence should still be checked on Sunshine's own product variants. Tables and frozen results retain the words, excluded titles and counts.

A stronger scope challenge adds positive title words: hair dryer/hairdryer/blow dryer/blowdryer; microwave/microwaves; and pacifier/pacifiers/paci/soothie/soothies. It then excludes the declared accessory words. Genuine hood dryers are retained even if the title says stand; pure holders are excluded. This conservative lexical rule can miss genuine products and include combined packages, so it is a target-title sensitivity rather than certified product labels.

**Table 4.** A stricter positive-title cohort for transfer to the three intended products.

| Category | Target reviews | Families | Mean stars | Low-star |
|---|---|---|---|---|
| Hair dryer | 11035 | 123 | 4.122 | 14.4% |
| Microwave | 1467 | 39 | 3.337 | 34.4% |
| Pacifier | 10475 | 286 | 4.419 | 9.1% |

The pacifier file is particularly heterogeneous: only 10,475 of 18,939 rows meet this narrow scope, and their low-star fraction is 9.1% instead of 11.3%. Target-cohort design comparisons below use that explicit membership. This scope correction was made after the first complete paper and original final-test results were inspected, in response to the observed title/concordance mismatch. Its predictive comparison is transparently secondary; it does not restore an untouched test set.

### 3.2 Helpfulness is conditional on receiving votes

For reviews receiving at least one vote, helpfulness is the fraction of votes marked helpful, $u_i = h_i/v_i$. We report this fraction together with vote coverage, $\Pr(v_i > 0)$, because reviews without votes have no observed helpfulness judgment. The aggregate $\sum_i h_i / \sum_i v_i$ weights reviews in proportion to their vote counts; an average across reviews would answer a different question.

**Table 5.** Cumulative voting and text length.

| Category | Helpful/votes | Age–votes $\rho$ | Length–votes $\rho$ |
|---|---|---|---|
| Hair dryer | 85.0% | 0.31 | 0.37 |
| Microwave | 84.3% | 0.45 | 0.46 |
| Pacifier | 73.1% | 0.23 | 0.36 |

Vote coverage is 66.9% for microwave reviews, 37.7% for hair dryers and 28.0% for pacifiers. Many pacifier and hair-dryer reviews therefore provide no measured helpfulness judgment. Zero votes must not be interpreted as zero usefulness. Time since the latest recorded review is positively correlated with total votes, as is text length. Length–vote correlations remain positive within age quartiles in all categories. This supports an exposure-aware presentation, but longer writing could also reflect severe problems, product complexity or selection.

For launch monitoring, Sunshine should show issue prevalence among *all* reviews and helpfulness only among voted reviews, always displaying both denominators. A vivid, heavily voted complaint warrants examination, but weighting every reputation estimate by votes would confound product quality with exposure and aging. This is why our forecasting model deliberately uses no helpfulness field.

**Table 6.** Extreme ratings and vote reception in the supplied files.

| Category | 1-star voted | 5-star voted | 1-star helpful | 5-star helpful |
|---|---|---|---|---|
| Hair dryer | 67.9% | 31.7% | 78.2% | 89.0% |
| Microwave | 79.1% | 56.8% | 80.0% | 90.2% |
| Pacifier | 64.3% | 21.5% | 61.3% | 79.8% |

One-star reviews are more likely than five-star reviews to receive any votes, yet their conditional helpful-vote fractions are lower. Their text is also longer on average in all three files. This distinguishes attention from endorsement: a complaint can attract votes without every voter endorsing it. Neither difference certifies a true defect or proves that readers change purchases. Age, listing and reviewer selection are unresolved.

## 4 Text descriptors and product-feature associations

We normalize case, replace HTML markup with spaces, and use letter tokens from headline plus body. Enthusiasm is operationalized by love/loves/loved, excellent, perfect, great

and amazing; recommendation is tracked separately and also enters the positive bundle. Disappointment includes disappointed/disappointing/disappointment, terrible/horrible/awful, broken/broke/breaks/defective, stopped and return/refund variants. A positive match is suppressed if a negator occurs among the preceding three tokens, including cannot and contracted would-not/could-not forms. Curly apostrophes are normalized. Recommendation intensification such as cannot recommend enough is retained only without a following not/no/never; the contracted expression "couldn't be more perfect" is retained. These are scoped heuristics rather than complete sentiment parsing. This dictionary is declared in executable source; no external sentiment corpus or pretrained sentiment labeler is used.

## 4.1 Descriptor associations with rating levels

To compare rating patterns in reviews with and without a descriptor, let $a$, $b$ count low/non-low reviews with a mention and $c$, $d$ those without. The odds ratio compares the low-star odds in these two groups. A half-count correction keeps the ratio defined when a contingency cell is empty:

$$O = \frac{(a + 1/2)(d + 1/2)}{(b + 1/2)(c + 1/2)}. \tag{1}$$

An approximate log-odds interval is $\log O \pm 1.96\sqrt{V}$, where

$$V = \sum_{z \in \{a,b,c,d\}} \frac{1}{z + 1/2}. \tag{2}$$

An odds ratio above one indicates higher corrected low-star odds among mentioning reviews. The interval describes count-based association and ignores within-family clustering. Fisher tests and Benjamini–Hochberg adjusted values for all 33 declared category–descriptor comparisons are retained in the results; they are exploratory under the usual dependence conditions, not product-effect tests.

Expressions of love and perfection associate with higher ratings, whereas disappointment, breakage and returns associate with low ratings in each supplied category. The full one–five-star distributions are retained in the machine-readable results, so the analysis does not force all text into two sentiments. The pattern answers question 2(e), but does not prove that changing review wording changes the star rating. Stars and text are jointly produced by an author's experience, and their disagreement is itself useful for triage.

We inspect deterministic concordances for broken, disappointed, perfect and recommend: three matching review IDs per category and descriptor. These examples expose ambiguity, such as praising a replacement while mentioning an earlier break or saying a feature is perfect for only one use. They are neither a random accuracy sample nor human labels. The rule distinguishes would-not/cannot-recommend wording from cannot-recommend-enough intensification and the tested enough-not-to-buy adverse exception. It does not implement general postposed negation. Six targeted source-text cases check these local boundaries; they do not establish global semantic accuracy. Earlier rule versions and their results remain in the evidence archive. The dictionary therefore remains a measured *mention* system. A customer-service analyst should read flagged text before identifying a root cause.

**Table 7.** Declared descriptors and low-star associations. OR compares a mention with no mention.

| Cat. | Descriptor | $n$ | Low-star | OR | 95% interval |
|---|---|---|---|---|---|
| H | love | 2668 | 3.9% | 0.19 | [0.16, 0.23] |
| H | perfect | 617 | 3.6% | 0.21 | [0.14, 0.32] |
| H | disappointed | 345 | 57.1% | 8.71 | [6.99, 10.85] |
| H | broken | 434 | 48.6% | 6.20 | [5.10, 7.55] |
| H | return | 447 | 68.9% | 15.67 | [12.73, 19.29] |
| M | love | 183 | 13.1% | 0.30 | [0.19, 0.46] |
| M | perfect | 155 | 3.2% | 0.07 | [0.03, 0.16] |
| M | disappointed | 49 | 65.3% | 4.17 | [2.31, 7.54] |
| M | broken | 105 | 82.9% | 11.99 | [7.17, 20.04] |
| M | return | 99 | 70.7% | 5.77 | [3.70, 8.99] |
| P | love | 6846 | 2.8% | 0.15 | [0.13, 0.18] |
| P | perfect | 1453 | 1.6% | 0.12 | [0.08, 0.18] |
| P | disappointed | 400 | 59.8% | 13.00 | [10.59, 15.96] |
| P | broken | 237 | 50.6% | 8.48 | [6.55, 10.98] |
| P | return | 312 | 67.0% | 17.53 | [13.79, 22.29] |

## 4.2 Design topics and family-adjusted comparisons

Each target-title cohort has a small declared topic dictionary. Reliability uses words such as broke, broken, stopped or defective; dryer topics also include noise, weight, heat and power/speed. Microwave topics include size/capacity, heating, noise and controls. Pacifier topics include acceptance, cleaning, size/fit, material and attachment. Positive and negative topic words are both retained: a "quiet" mention is not a noise complaint.

Here OR compares the low-star odds with versus without a topic. "Within" is a family-adjusted risk difference: for families having at least three mentioning and three non-mentioning reviews, calculate the difference in their low-star fractions and average it with weights $n_1 n_0/(n_1+n_0)$. This reduces stable listing-mix confounding without making a causal claim. The number of eligible families is reported to expose sparse comparisons.

Reliability mentions have large negative-review associations for dryers and microwaves, and these persist within families. To check concentration, we delete each originally eligible family in turn and recalculate the weighted difference, keeping the measurement and remaining-family eligibility fixed. Table 9 is a deterministic influence range, not a confidence interval or a guarantee against all confounding. Every deletion keeps the electrical-product association positive; no single eligible family alone explains its direction. The pacifier material comparison remains close to zero and its deletion range spans zero.

We therefore most confidently recommend prioritizing durability investigation for both electrical products: inspect returned units, separate early failures from later wear, and validate control and motor/heating-system endurance before promoting extra features. The data identify complaint themes, not the physical failure mechanism or a quantified return on investment.

**Table 8.** Target-title issue mentions. Within-family differences are percentage points.

| Cat. | Mention | *n* | Low-star | OR | Within | Families |
|---|---|---|---|---|---|---|
| H | reliability | 959 | 51.3% | 8.58 | +38.1 | 81 |
| H | noise | 1136 | 14.6% | 1.02 | +0.2 | 89 |
| H | weight | 1932 | 9.7% | 0.59 | -5.5 | 102 |
| H | heat | 2307 | 17.2% | 1.31 | +3.8 | 114 |
| H | power/speed | 3275 | 8.1% | 0.43 | -8.3 | 120 |
| M | reliability | 255 | 80.0% | 12.00 | +41.1 | 33 |
| M | size/capacity | 480 | 17.1% | 0.28 | -17.3 | 24 |
| M | heating | 394 | 34.5% | 1.01 | -2.3 | 34 |
| M | noise | 142 | 32.4% | 0.91 | -6.2 | 16 |
| M | controls | 295 | 35.6% | 1.07 | -0.9 | 30 |
| P | acceptance | 1465 | 9.8% | 1.10 | -0.1 | 134 |
| P | cleaning | 957 | 7.3% | 0.78 | -0.8 | 75 |
| P | size/fit | 2069 | 11.8% | 1.47 | +3.2 | 149 |
| P | material | 548 | 14.8% | 1.81 | +0.2 | 62 |
| P | attachment | 494 | 9.1% | 1.01 | +0.4 | 42 |

**Table 9.** Deleting each eligible family once.

| Topic | Families | Within (pp) | Deletion range (pp) | Positive families |
|---|---|---|---|---|
| H: reliability | 81 | +38.07 | [+36.96, +39.00] | 76/81 |
| M: reliability | 33 | +41.10 | [+38.17, +42.94] | 29/33 |
| P: material | 62 | +0.23 | [-0.94, +1.16] | 23/62 |

The near-zero within-family association for the dryer noise topic may conceal opposing evaluations, because that topic groups noise and quiet expressions together. To examine this possibility, we compare target-cohort reviews containing noisy/noise/loud but not quiet with reviews containing quiet but none of those three words. Eligible families must have at least three reviews in each group; the comparison uses the same family weights as the topic analysis. Reviews mentioning both sides are excluded from this contrast: 101 dryer reviews, of which 16 are low-star, and five microwave reviews, of which two are low-star. These mixed mentions remain in the recorded counts.

**Table 10.** Noise and quiet expressions in the target cohort.

| Rule | Noise $k/n$ | Quiet $k/n$ | Families | Within (pp) |
|---|---|---|---|---|
| H: literal | 122/581 | 28/454 | 26 | +7.89 |
| H: negation | 117/428 | 26/439 | 24 | +13.29 |
| M: literal | 33/87 | 3/26 | 1 | +21.74 |

The table's $k/n$ values use all target reviews in each lexical group; its within-family difference uses only eligible paired families. For the literal dryer comparison these paired counts are 41/300 versus 18/330 across 26 families. The difference is 7.89 percentage points and ranges from 6.93 to 9.13 after deleting each family. Suppressing each hit with a declared negator in its preceding three tokens gives 13.29 points across 24 families, with deletion range 11.99–15.12. These are different rule-defined comparisons, not competing estimates of a human-labeled complaint effect.

Context explains the caution. Dryer review R10BDCAM5M8MEG carries one star and says "very noisy"; R1008ZQFTNURF1 carries four stars and describes a "nice quiet hair dryer". Yet five-star R117LX05NTTAZR says "not obnoxiously loud", and five-star R11NN1XLQT605W says "not too loud" and "not exactly quiet". Literal topic membership cannot safely determine the feature's evaluation. The local-negation rule also cannot resolve every comparison, replacement product or word sense.

Only one microwave family qualifies for the literal noise–quiet comparison (parent 423421857: 5/23 versus 0/9). Removing it leaves no estimate. Moreover the original microwave noise topic also includes beep/beeps, whereas this new comparison does not. The 21.74-point contrast therefore cannot establish a general microwave design preference. For dryers, the combined noise topic's near-zero association should not be read as indifference to noise: opposing expressions were mixed. Read the implicated contexts and test noise-related complaints alongside reliability, rather than promise a causal benefit from making every product quieter.

For pacifiers, the material-topic pooled association largely disappears within families, showing how a plausible-looking aggregate result can be driven by listing composition. Acceptance words also mix a baby's refusal with successful acceptance. We do not claim that one material or pacifier shape is universally preferred. The usable recommendation is to present clear age/size and cleaning information and test acceptance on the intended age range; use actual text to distinguish rejection, fit, cleaning and accessory issues. For dryers, power/speed and weight mentions often occur in favorable reviews; preserve portability and quick drying

while investigating reliability. For microwaves, size/capacity is frequently discussed favorably, so a small footprint alone should not be read as a defect: communicate actual usable capacity and plate compatibility.

# 5 Monitoring reviewer reputation

## 5.1 Posterior risk estimates and investigation thresholds

At cutoff $t$, collect reviews in $[t-90, t)$ using real calendar days. Track $n$, $k/n$, mean stars and the histogram; positive/negative and reliability-mention fractions; verified-review strata; vote coverage and conditional helpfulness separately. The window highlights current feedback while cumulative averages serve as context. A new listing with few reviews is visibly uncertain rather than automatically praised or condemned.

For current low-star risk, use the category's pre-2014 low-star mean $p_0$ as a disclosed reference and strength $s = 10$:

$$p \mid k, n \sim \text{Beta}(k + sp_0,\ n - k + s(1 - p_0)), \qquad E[p \mid k, n] = \frac{k + sp_0}{n + s}. \tag{3}$$

This follows by multiplying the binomial likelihood $p^k(1-p)^{n-k}$ by the Beta prior density. The 95% interval is given by its 2.5% and 97.5% quantiles, computed using SciPy [3]. The interval quantifies uncertainty inside this model, including its fixed prior choice, rather than selection bias or variant mixing.

An *investigation alert* requires at least ten recent reviews and

$$\Pr(p > p_0 + 0.10 \mid k, n) \geq 0.95. \tag{4}$$

The ten-percentage-point margin is an illustrative managerial tolerance. It is not optimized from unobserved business costs. Alerts prompt reading complaints and inspecting batches; they do not alone authorize withdrawing a product. For each alert, the saved results repeat its probability with strengths two and thirty. A decision that changes under these values receives the label "prior-sensitive" and requests more evidence.

**Table 11.** Target-title monitoring at August 31, 2015; alerts are relative to a historical reference.

| Category | Baseline low | Eligible families | Alerts |
|---|---|---|---|
| Hair dryer | 17.2% | 30 | 0 |
| Microwave | 44.1% | 3 | 0 |
| Pacifier | 9.8% | 35 | 0 |

The final dashboard uses the target-title cohort over June 3–August 31, 2015, inclusive. No eligible family meets the 95% relative-alert rule, and this result is unchanged at prior strengths two and thirty. The absence of an alert indicates insufficient evidence of a ten-percentage-point excess above the historical reference. It does not certify acceptable product quality: the microwave reference itself has 44.1% low-star reviewing.

A relative change threshold and an acceptable risk threshold serve different purposes. Before deployment, management must choose an acceptable reviewer-risk tolerance $u$ independently of the competitor reference. If $A$ is the cost of a false investigation and $D$ the cost of missing a genuinely excessive-risk product, the expected losses are $A(1 - P)$ for investigating and $DP$ for deferring, where $P = \Pr(p > u \mid k, n)$. Investigation has lower expected loss when $P > A/(A + D)$. The supplied files contain neither $u$ nor these costs, so the illustrative 95% rule cannot be interpreted as a cost-optimal decision policy.

For a new Sunshine listing, the competitor prior is provisional and should be replaced as the company acquires its own review history. Counts and genuine complaints must remain visible during this transition. A historically poor reference cannot make an unacceptable reviewer-risk level acceptable.

Individual complaints can also warrant an engineering investigation before a frequency threshold is crossed. The statistical alert summarizes repeated reviewing outcomes; it does not replace examination of a specific incident.

**A worked investigation decision.** The target-cohort competitor family 565072108 has five low-star reviews among eleven reviews in the last window. With $p_0 = 0.441489$ and $s = 10$, its posterior is $\text{Beta}(9.414894, 11.585106)$. The probability of exceeding $p_0 + 0.10$ is 0.19492, so the relative 95% alarm is off. For an explicitly illustrative policy with acceptable reviewer-risk $u = 0.25$ and loss units $A = 1, D = 4$, however, $\Pr(p > u) = 0.97333 > 0.20 = A/(A + D)$. Expected loss is 0.02667 for investigating and 3.89331 for deferring under this binary loss model. Thus this example warrants investigation even without a relative alert. The counts are observed competitor data; the tolerance and costs are hypothetical, not Sunshine measurements or defect probabilities.

Operationally, the analyst opens the five low-star source reviews, verifies product/variant and intended-use context, and routes repeated failure descriptions to the appropriate engineering test. Record the evidence and test result before deciding any redesign. At the next fixed calendar update, recompute the same counts and risk rule; only management's declared acceptance policy and investigation evidence can close a case. A falling cumulative average or absence of a relative alarm alone cannot close it.

## 5.2 Calendar trends and composition effects

A fall in $q$ or rise in mean stars suggests improving reviewer reputation; rising issue prevalence or a low-star burst suggests deterioration. Growth in the number of reviews is ambiguous: it may accompany popular sales, a review campaign or controversy. Figure 2 consequently plots reputation proportions rather than treating the volume trend as success.

We compare full-year 2014 with January–August 2015. To distinguish changing products from changing responses, restrict to families with at least twenty reviews in each period and standardize both periods using their 2014 review counts:

$$Q_{\text{old}} = \frac{\sum w_j q_{j,\text{old}}}{\sum w_j}, \qquad Q_{\text{new}} = \frac{\sum w_j q_{j,\text{new}}}{\sum w_j}, \quad w_j = n_{j,\text{old}}. \tag{5}$$

The family weights are fixed for both quantities; results cannot be explained simply by larger later weights on popular families. This comparison still mixes seasonal differences and limits

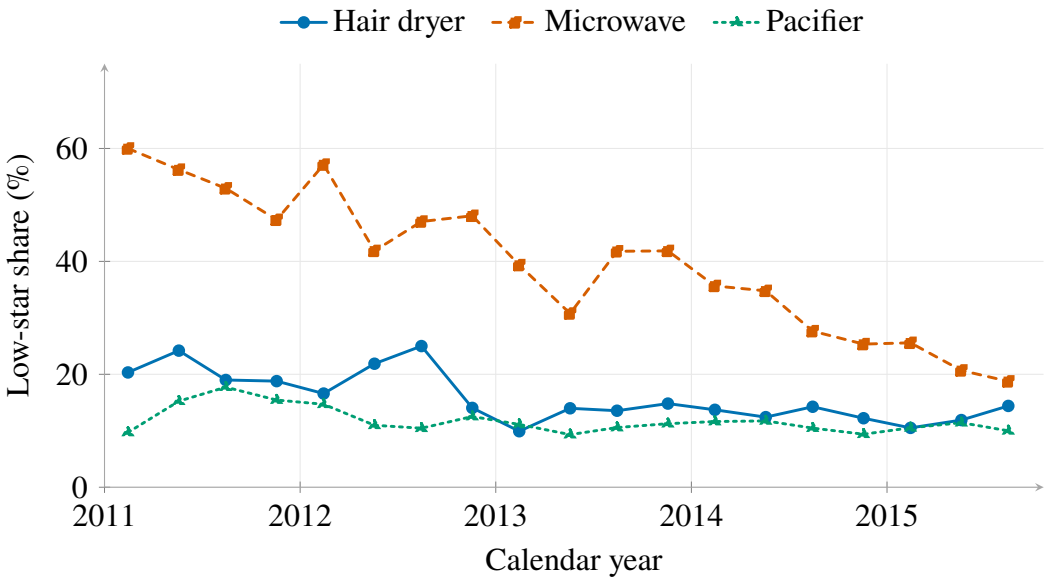



**Figure 2.** Quarterly reviewer risk. Points summarize calendar quarters with at least ten reviews; 2015 Q3 stops on August 31. Listing composition may change between points.

itself to established families, so it is not a universal annual trend.

**Table 12.** Pooled and fixed-composition reputation changes.

| Category | 2014 low | 2015 low | Families | Fixed 2014 | Fixed 2015 |
|---|---|---|---|---|---|
| Hair dryer | 13.1% | 11.9% | 28 | 12.0% | 11.0% |
| Microwave | 30.3% | 22.0% | 3 | 16.0% | 13.9% |
| Pacifier | 10.5% | 10.7% | 33 | 7.3% | 9.6% |

The pooled microwave low-star share declines much more than its matched-family standardized share. The pacifier matched-family risk increases even though its pooled fraction changes little. These are useful cautions against reading a file-wide average as a product's own reputation. The microwave matched subset includes an accessory; the exact family list is retained, and the title sensitivity should be applied before transferring the comparison to an oven launch.

**Table 13.** Fixed-composition changes inside the target-title cohort.

| Category | 2014 low | 2015 low | Families | Fixed 2014 | Fixed 2015 |
|---|---|---|---|---|---|
| Hair dryer | 13.0% | 11.7% | 28 | 12.0% | 11.0% |
| Microwave | 33.4% | 24.1% | 2 | 16.8% | 16.0% |
| Pacifier | 8.2% | 9.2% | 30 | 6.6% | 9.3% |

Within the stricter cohort the established microwave comparison has only two families and changes from 16.8% to 16.0%, while the pooled cohort changes from 33.4% to 24.1%. Pacifier fixed-composition risk rises from 6.6% to 9.3%. These differences demonstrate the practical need to track an actual listing and product scope, while avoiding a general causal explanation of the temporal change.

**Table 14.** Selected family changes (percentage points); not discovery tests.

| Category | Parent ID | Old/new $n$ | Change | Posterior interval |
|---|---|---|---|---|
| Hair dryer | 195677102 | 23/102 | +7.8 | [-7.1, +11.4] |
| Microwave | 423421857 | 126/259 | +0.7 | [-8.4, +7.3] |
| Pacifier | 277493012 | 20/33 | +18.2 | [-0.7, +26.2] |

Intervals for selected family changes are computed by subtracting independent draws from the two Beta posteriors (10,000 draws, fixed seed 2020). They include substantial uncertainty; selecting the largest observed change does not make it statistically confirmed. For example, an apparent pacifier-family deterioration merits a closer read but is not evidence of a production change. Sunshine should monitor its own family and variant history, review growth, rating tail and topic counts together.

# 6 Predicting future low-star reviews

## 6.1 Prediction target and temporal evaluation

To answer 2(c), define success in the limited sense of *favorable future reviewer reputation*, and failure risk as the probability that a future review is one or two stars. At each calendar-quarter cutoff from January 2010 through April 2015, use the preceding 90 days for predictors and the following 90 days for outcomes. Prediction windows require at least five reviews on each side; the forecast therefore concerns active reviewed families. The broader flow analysis in Section 7 retains zero future counts.

Training cutoffs end in 2013; all their outcomes precede 2014. The 2014 cutoffs select model complexity and the model family. Final evaluation uses January 1 and April 1, 2015, whose complete future windows end before the supplied cutoff. No partial third-quarter outcome is invented. The fixed 90-day windows leave small gaps between adjacent quarters instead of pretending every quarter has ninety days. Product identity may repeat across time, which is appropriate for updating an existing listing; entirely new-product launch transfer is a different problem.

Let $m$ and $y$ be the future review count and its low-star count. A constant category model estimates its low-star rate from training outcomes. A recent-risk baseline predicts $(k+10p_0)/(n+10)$, where this forecast prior uses the training *eligible-window* outcome mean. Its different sampling denominator from the dashboard prior is explicit. We then compare three regularized logistic candidates [2]:

- Ratings: recent low-star fraction, scaled mean $(\bar{r}-3)/2$, $\log(1+n)/5$, and category indicators.
- Text: negative, positive and defect/return mention fractions, the same recent-count transform, and category indicators.
- Combined: all rating and text predictors with the same count and category terms.

"Text-only" here means no rating predictors; the shared volume and category terms keep structural information equal. The defect bundle is overlapping with negative text by design, so coefficient signs cannot be read as independent effects. We do not include vote snapshots, future review dates, or learned text parameters from the final test.

## 6.2 Model estimation and predictive performance

For predictor vector $x$, $p(x) = 1/(1+\exp(-a-x^T b))$. Fit the binomial log likelihood with an $\ell_2$ penalty on $b$; the intercept is unpenalized:

$$\min_{a,b} \; -\sum \{y \log p + (m-y)\log(1-p)\} + \frac{\|b\|^2}{2C}. \tag{6}$$

We implement each window as two weighted observations, with weights $y$ and $m-y$, which gives this exact aggregate likelihood. Each candidate selects $C \in \{0.1, 1, 10\}$ by 2014 review-weighted log loss; all selected $C$ values equal 0.1. Their parameters remain fitted to the training prefix: no final-test refit or threshold tuning is performed.

The main loss is

$$L = -\frac{\sum\{y \log p + (m-y)\log(1-p)\}}{\sum m}; \qquad B = \frac{\sum\{y(1-p)^2 + (m-y)p^2\}}{\sum m}. \tag{7}$$

The Brier loss $B$ provides a second probability score. Both operate at the future-review level and preserve the binomial denominator; a tiny window is not given the same precision as a large one. This choice favors high-volume families, so the results should not be represented as an equal-weight guarantee for every listing.

**Table 15.** Prediction comparison on identical calendar windows. Lower loss is better.

| Model | 2014 log loss | 2015 log loss | 2015 Brier |
|---|---|---|---|
| Category | 0.34352 | 0.34850 | 0.10000 |
| Recent | 0.33200 | 0.33129 | 0.09140 |
| Ratings | 0.32722 | 0.32402 | 0.09008 |
| Text | 0.33223 | 0.32969 | 0.09217 |
| Combined | 0.32685 | 0.32368 | 0.09012 |

The final test contains 3901 future reviews in 222 windows from 138 families; development contains 262 eligible windows. The combined model has the lowest log loss on both development and final-test data, but its final-test gain over ratings alone is 0.000342 nats per review. Ratings alone has a slightly lower final-test Brier loss. The practical importance of the combined model therefore depends on whether this small difference persists beyond the observed test sample.

Microwave validation rests on eight final-test families and 259 reviews. Interpretation is limited by the small microwave sample and the observed distribution shift. Pacifier and dryer samples are larger, but their active-family restriction still prevents unqualified transfer to a brand-new listing. Of 138 test families, 84 do not occur in the training snapshots, yet each supplies at least five recent reviews at prediction time; this is not a zero-history cold-start experiment.

**Table 16.** Final-test losses by category. Combined was selected on 2014 data.

| Category | Reviews | Families | Ratings | Text | Combined |
|---|---|---|---|---|---|
| Hair dryer | 1892 | 63 | 0.32598 | 0.33107 | 0.32636 |
| Microwave | 259 | 8 | 0.47397 | 0.53170 | 0.47922 |
| Pacifier | 1750 | 67 | 0.29971 | 0.29829 | 0.29777 |

We resample entire test families 1,000 times, retaining all their windows, and recompute the review-weighted loss difference. Intervals below are percentile product-cluster bootstrap intervals conditional on this test sample and fixed fitted models. They address clustering of losses, not model selection uncertainty or selection of reviewed families.

**Table 17.** Combined-model improvement in final-test log loss.

| Comparator | Gain | 95% product-cluster interval |
|---|---|---|
| Category | 0.024817 | [0.007557, 0.049701] |
| Recent | 0.007607 | [0.001813, 0.013766] |
| Ratings | 0.000342 | [-0.000597, 0.001410] |
| Text | 0.006006 | [0.002083, 0.010725] |

The product-cluster intervals support an improvement over the category and recent-risk baselines in this test sample. They do not establish an incremental advantage over ratings alone, whose interval includes zero. A rating-based warning model is therefore sufficient for the proposed baseline; text can still identify the complaint themes that analysts need to investigate.

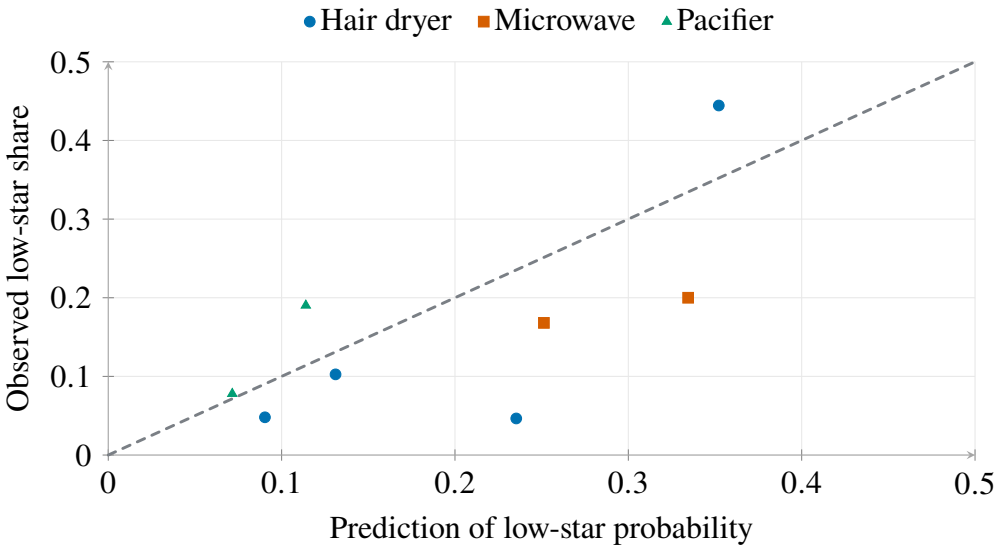



**Figure 3.** Held-out calibration. Points use fixed risk bins and review-weighted means from the combined model; the diagonal denotes agreement. Some bins have few reviews, so this is a diagnostic rather than a calibration guarantee.

Figure 3 shows imperfect calibration, especially for microwaves and sparse upper-risk bins. We therefore do not translate forecast probabilities directly into an automatic launch/no-launch

decision. A warning score may rank investigation needs, but financial costs, recall thresholds, sales response and a prospective calibration test are absent. Rechecking these on Sunshine's own reviewed products is a condition for deployment.

## 6.3 Sensitivity to product scope and review coverage

The stricter title cohort repeats every candidate with the same predictors, 90-day calendar definition, training years and 2014 selection rule. It has 3771 test reviews in 211 windows and 131 families.

**Table 18.** Secondary target-cohort sensitivity with the same algorithms and calendar split.

| Model | 2014 log loss | 2015 log loss | 2015 Brier |
|---|---|---|---|
| Category | 0.33551 | 0.34486 | 0.09845 |
| Recent | 0.32606 | 0.33032 | 0.09085 |
| Ratings | 0.32120 | 0.32352 | 0.08985 |
| Text | 0.32562 | 0.32759 | 0.09135 |
| Combined | 0.32081 | 0.32308 | 0.08986 |

The combined model again wins nominally, but its gain over ratings is 0.000437 with a product-cluster interval [-0.000535, 0.001544]. The conclusion is unchanged: text is valuable for complaint routing, while its incremental forecast improvement is not established. Because this cohort definition followed inspection of the original test and concordances, these numbers are a scope robustness check rather than a second untouched confirmation.

**Coverage at prediction time.** A deployed dashboard must determine eligibility from past reviews alone. The primary evaluation additionally requires at least five future reviews, which restricts the population on which its loss is measured. We therefore retain that comparison and assess all 300 past-eligible windows in 2015: 298 have observed future reviews and 2 have none. Predictions are available for the latter two windows, but their observed low-star risks are undefined. The primary comparison covers 74.0% of the past-eligible windows; no future value enters its predictor vector.

**Table 19.** Secondary coverage check: every 2015 window eligible from its past.

| Model | Primary loss | All positive loss | Small-future loss |
|---|---|---|---|
| Category | 0.34850 | 0.35607 | 0.49159 |
| Recent | 0.33129 | 0.33832 | 0.46419 |
| Ratings | 0.32402 | 0.33154 | 0.46599 |
| Text | 0.32969 | 0.33702 | 0.46829 |
| Combined | 0.32368 | 0.33104 | 0.46272 |

The 298 positive-count windows contain 4119 reviews. The small-future windows omitted from the primary comparison contribute 218 reviews, including 39 low-star reviews (17.9%), and have higher predictive losses than the reviewer-rich subset. On all positive-count windows,

the combined model's log loss is 0.33104, compared with 0.33154 for ratings alone. This coverage analysis was added after the primary test had been inspected; the original training and development eligibility rules remain unchanged. The revised historical comparisons also incorporate the disclosed lexical repair. These are secondary analyses, and the separately frozen first complete draft retains its original results.

# 7 Prior ratings and subsequent review volume

A sequence of poor ratings could attract complaint reviews, discourage purchases and hence reduce reviews, or reflect an underlying product problem that affects both earlier and later authors. The files observe neither what prospective customers read nor the non-reviewers. We first test an association that the available information can actually support. The panel below uses the preceding low-star share; it does not estimate separate associations for every star level or for specific rating sequences.

For every eligible past-quarter window with at least five recent reviews, retain the next ninety-day review count $m$, including $m = 0$. Restrict to families with at least three such windows so a within-family comparison exists. Estimate separately in each category:

$$\log(1 + m) = \alpha_j + \tau_t + \beta q + \gamma \log(1 + n) + e. \tag{8}$$

Family effects $\alpha_j$ adjust persistent differences; calendar-cutoff effects $\tau_t$ adjust common shocks; prior count controls review-flow persistence. Least squares gives a transparent descriptive benchmark. Standard errors cluster by family, with a finite-sample correction. The intervals use a normal approximation; eight microwave families are too few for strong inferential assurance.

**Table 20.** Past low-star share and subsequent review flow.

| Category | Windows | Families | $\beta$ | Approx. 95% interval |
|---|---|---|---|---|
| Hair dryer | 461 | 63 | -0.159 | [-0.634, 0.316] |
| Microwave | 50 | 8 | -0.055 | [-1.060, 0.949] |
| Pacifier | 353 | 58 | -0.134 | [-0.712, 0.445] |

All three coefficient intervals cross zero. Repeating the panel on the target-title subset gives dryer $\beta = -0.159$, microwave $-0.133$ and pacifier $-0.096$; all three intervals still cross zero. The target microwave comparison has only seven families. This scope check does not remove unmeasured exposure or selection. We find no clear adjusted association between prior low-star share and subsequent review volume in these eligible windows. A ten-point rise in $q$ corresponds to $\exp(0.1\beta) - 1$ in the fitted $1 + m$ scale, not a causal percent change in purchases. Including zero-count future windows avoids selecting only products that continued attracting reviews.

The rank correlations between prior and subsequent low-star proportions are positive (approximately 0.12 for dryers, 0.52 for microwaves and 0.37 for pacifiers among positive-count windows). That is consistent with persistent dissatisfied experiences, reviewer selection or con-

tagion, and cannot distinguish them. It provides a reason to inspect repeat complaints, not to claim that preceding reviewers persuaded later authors to complain.

**Why causal incitement is not identified.** Even in the simple expectation $E[m] = S\pi$, sales $S$ and review propensity $\pi$ are both unobserved. Counts changing from one to two can arise with sales changing from ten to twenty at unchanged $\pi = 0.1$, or unchanged sales of one hundred with $\pi$ changing from 0.01 to 0.02. Both worlds can have the same prior stars and observed subsequent reviews. A further unobserved variable can determine which prior stars were displayed. Thus a fixed-effect fit cannot identify how seeing ratings changes reviewing behavior.

Sunshine should not solicit a string of particular ratings to stimulate review volume on the basis of these files. To test incitement, it would need a separately authorized design recording impressions, displayed rating histories, purchases and review opportunities, with random or defensible quasi-random variation. Such new data are outside this problem; the current answer is a tested association plus a constructive identification limit, rather than a causal yes/no invented from correlations.

# 8 Implications for product design and monitoring

Our most confident recommendation is a reliability-first investigation program for dryers and microwaves. It is supported by large low-star associations, persistence within product families, and a sensitivity that excludes explicit accessory titles. This evidence ranks an investigation priority; it does not measure how much revenue a durability redesign will earn. The engineering team should code actual failure modes from returned units and complaints before deciding which component to alter.

For the hair dryer, preserve the favorably described quick drying and manageable weight, communicate intended use, and monitor repeated stoppage/breakage alongside heat-related text. The merged noise topic masks opposing expressions; separately read noise and quiet contexts and investigate repeated acoustic complaints. The reliability signal remains supported across many families, whereas one eligible microwave noise comparison is insufficient for a general design claim. For the microwave, preserve clearly communicated usable capacity and plate fit while auditing early failures and controls; do not interpret favorable size discussion as proof that every buyer wants a larger oven. For the pacifier, avoid deriving a universal material preference from the pooled data. Separate core pacifiers from clips, cases, novelty items and unrelated baby goods in Sunshine's dashboard, publish size/age and cleaning information, and test the intended use with a correctly scoped product cohort.

After launch, use the past ninety-day dashboard as a compact weekly view, updating counts without changing its calendar definition. Show the low-star posterior with its interval, actual issue count, verified strata, review volume and time since launch. A new listing starts as "insufficient own history." An alert leads to a named product-family investigation, variant check and review of new batches; a cleared investigation remains in the record rather than disappearing because a cumulative mean recovered.

The predictive rating baseline can supplement this process once at least five recent reviews exist. Text remains valuable because it names why a review may be unfavorable, even where its

incremental forecast loss is indistinguishable from zero. Display helpfulness as reception of already visible reviews, with vote coverage and age context, rather than letting a highly voted early review dominate a supposed quality score. The recommended workflow has observable inputs and an explicit action boundary; it requires neither unobserved sales nor fabricated customer histories.

# 9 Computational verification and study limitations

Independent calculations check the main numerical results at the level where they are used. Raw-record reconciliation checks review populations and vote totals; independent contingency counts check descriptor associations; exact-rational family contrasts and one-family deletion check the reported associations and their concentration. Posterior-density integration checks monitoring probabilities, raw-calendar rescanning and rowwise scoring check forecast results, and residualization checks the review-flow coefficient. The original analysis and the later lexical/family-influence supplement are separately versioned in the accompanying archive.

The verifier follows different computational paths from the model. It uses standard-library parsing rather than pandas summaries, exact cross-products rather than log-odds evaluation, density integration rather than a distribution survival call, and Frisch–Waugh–Lovell residualization rather than the coefficient returned by the full fitted panel. These comparisons can detect implementation or numerical discrepancies while retaining common data and measurement definitions.

Agreement between these calculations does not validate the meanings assigned to words or remove selection bias. Both implementations can reproduce the same incorrect text interpretation. The checks therefore support the stated computational results; semantic accuracy and causal identification require different evidence.

Robustness checks have different purposes. Verified-only and category-level one–three-star summaries describe reviewer and label sensitivity without certifying all downstream robustness; title exclusions challenge category scope; within-family comparisons challenge listing mix; prior strengths challenge sparse-data monitoring; temporal holdout challenges a 90-day forecast; and product-cluster resampling challenges independence of forecast losses. No parameter perturbation alone certifies selection-free population quality. The evidence archive retains uncompetitive forecast candidates and failed attempts with their actual identities.

The principal unresolved quantities are sales and profit, exposure and selective reviewing, true defect incidence, latent sentiment accuracy, zero-history launch performance, and prospective calibration. A broad extracted category includes accessories, unrelated products and many short-lived families; a future Sunshine product is not exchangeable with all of them merely because its name is similar. The 2015 horizon is a historical test, not validation of later markets or a multi-year forecast. These limits restrict deployment while leaving the observed relationships and monitoring formulas useful.

# 10 Conclusion

The three files support an evidence-based launch dashboard and a concrete design investigation priority. Track dissatisfied reviewing with its count and uncertainty, retain the rating distribution and verified strata, and read reliability complaints before drawing engineering conclusions. Product and reviewer composition materially affect category averages. Combined text and ratings give a competitive historical warning baseline, but the observed incremental gain over ratings alone is too small and uncertain to justify a strong superiority claim. Prior low ratings do not show a clear adjusted review-volume association here, and their causal incitement effect is not identified. Sunshine can act on the measured complaint themes while collecting the additional evidence needed for sales, causality and prospective decisions.

# References

[1] COMAP. *2020 MCM Weekend 2, Problem C: A Wealth of Data*. Original three-page problem, glossary and supplied `hair_dryer.tsv`, `microwave.tsv`, `pacifier.tsv`. Data periods end August 31, 2015. All numerical product evidence derives from these supplied records.

[2] scikit-learn developers. *Linear Models: Logistic Regression*, version 1.9.1 documentation. https://scikit-learn.org/stable/modules/linear_model.html. Generic method/software documentation consulted October 12, 2026.

[3] SciPy developers. *scipy.stats.beta*, online manual. https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.beta.html. Consulted October 12, 2026; local execution version 1.18.1. Used for the explicitly derived Beta posterior quantiles and tails.

# A Reproducibility and full model parameters

The preserved run receipt records Python 3.12.0, pandas 2.3.3, NumPy 2.5.3, SciPy 1.18.1 and scikit-learn 1.9.1, the exact argv, raw-input hashes, code snapshot and output hashes. The case contains executable `code/model.py` and `code/validate.py`, saved predictions, every calendar snapshot, the complete descriptor/aspect tables and concordance candidates. The plotting and paper generator uses those saved results, not fresh model fits. No external product data or learned sentiment corpus enters the numerical analysis. The frozen autonomous version predates same-topic reference reading; this learning revision follows two such readings, recorded separately in the study archive.

The archive contains the original failed attempts, exact run receipt, all model coefficients and independent checks, together with the separately computed lexical/family-deletion supplement and its raw-CSV/Fraction verification. The three original data files, forecast fits, saved predictions, panel coefficients and figure arrays have not changed in this revision. The worked policy example additionally binds the observed family counts to an explicitly hypothetical tolerance and loss ratio.

A4 and 12pt are elected project house style, not asserted as official 2020 requirements. The original problem's twenty-page solution and twenty-four-page total limit are used; references and appendices appear after the solution and are excluded according to that problem. Historical general AI/font rules are not asserted as verified.

**Table 21.** Combined-model parameters fitted only on the training prefix.

| Term | Coefficient |
|---|---|
| Intercept | -0.96532 |
| risk | 0.30523 |
| mean scaled | -0.80545 |
| logn | -0.39433 |
| negative | 0.05887 |
| positive | -0.39416 |
| defect | -0.04124 |
| microwave | 0.68260 |
| pacifier | -0.49926 |

The recommended ratings-only baseline uses the same aggregate likelihood, training prefix and selected regularization $C = 0.1$. Its fitted parameters are listed separately for reproduction; no coefficients were refitted during this writing revision.

**Table 22.** Ratings-only warning-model parameters.

| Term | Coefficient |
|---|---|
| Intercept | -1.14304 |
| risk | 0.33475 |
| mean scaled | -0.90689 |
| logn | -0.39456 |
| microwave | 0.67575 |
| pacifier | -0.51770 |

# Report on Use of AI

This is an AI-authored historical-practice learning revision, with example team number 7391856 and no actual contest submission. The author AI formed definitions, wrote and executed model and validator source, inspected results, wrote the English paper and letter, generated vector figures and performed visual/code checks. The coordinating AI fixed the execution contract, input identity and budget before author work, then independently inspected the frozen first complete draft and returned formatting, prediction-scope and lexical counterexample findings. These findings led to repair of long formula/hash lines and appendix/AI page boundaries, a scoped negation repair, and an all-past-eligible evaluation. All revised historical checks disclose their post-first-test identity. Any final qualitative evaluation belongs to the accompanying review record; no award is claimed by the author. No human participant's independent mathematical work or semantic verification is claimed.

The accessible runtime identifies this service as Codex based on GPT-6; an exact provider model identifier and actual model token/cost metering are unavailable and remain unknown. The transcript provided to this task and retained technical events are not a complete provider chat export. We do not fabricate a complete prompt/response log. Program stdout, stderr, timestamps, hashes and source snapshots are technical records, not full AI conversations.

The permitted generic Praxis resources and three frozen lessons were read before route selection: measure the actual proxy and selection mechanism; preserve real calendar intervals and a matching forecast horizon; and validate task performance rather than representation fit. These do not provide answers to this problem. No same-topic paper, author code or targeted human feedback was accessed before the separately frozen autonomous version. After that seal, two AI readers studied complete 2020C papers by teams 2009116 and 2010638, and an independent AI reviewer first assessed the baseline before seeing those references. The present learning revision then separated acoustic word directions, added exhaustive family-influence checks and clarified investigation actions. It is not a new first-autonomous trial; no human labeling or actual award is claimed. Reference labels and printed claims were critically checked rather than adopted as numerical truth. A subsequent scientific-writing revision clarified the recommendation evidence, paragraph connections and formula interpretation using published writing guidance. It retained the numerical results and the earlier frozen versions; the revision does not create new product evidence or human validation.